# THE CLASS MAP FROM THE PICARD GROUP TO THE WEIL DIVISOR CLASS GROUP IS INJECTIVE

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Carrier window 3: statement and proof both lie inside the window 6000-8000 granted to this lane by the owner lane lane-s13 on 2026-10-14.

*Editorial note.* The statement refers to the displayed map (equation-c1) of the chapter, which is resolved in the assembly as a navigation pointer.

*Editorial note.* The references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

*Editorial note.* Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

*Allowed standard prerequisites (editorial).*

- The divisor of a meromorphic section is well defined. Lemma lemma-divisor-meromorphic-well-defined, used at proof lines 6886-6888. Allows replacing the section by one with zero divisor.
- A normal local ring of height one is a discrete valuation ring. Algebra, Lemma algebra-lemma-criterion-normal, used at proof lines 6897-6899. Turns vanishing of the order into a unit condition.
- A normal domain is the intersection of its height one localisations. Algebra, Lemma algebra-lemma-normal-domain-intersection-localizations-height-1, used at proof lines 6901-6904. Concludes that t and t^{-1} lie in A.

*Source reference.* Original lemma 031S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3995–3995. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 031T: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3996–3996. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original equation 0BE7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14340–14340. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

*Source reference.* Original lemma 02SH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3660–3660. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

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*Date*: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

**Lemma 0.2.** *Let $X$ be a locally Noetherian integral scheme. If $X$ is normal, then the map (0BE7)* $\mathrm{Pic}(X) \to Cl(X)$ *is injective.*

*Proof.* Let $\mathcal{L}$ be an invertible $\mathcal{O}_X$-module whose associated Weil divisor class is trivial. Let $s$ be a regular meromorphic section of $\mathcal{L}$. The assumption means that $\mathrm{div}_{\mathcal{L}}(s) = \mathrm{div}(f)$ for some $f \in R(X)^*$. Then we see that $t = f^{-1}s$ is a regular meromorphic section of $\mathcal{L}$ with $\mathrm{div}_{\mathcal{L}}(t) = 0$, see Lemma 02SH. We will show that $t$ defines a trivialization of $\mathcal{L}$ which finishes the proof of the lemma. In order to prove this we may work locally on $X$. Hence we may assume that $X = \mathrm{Spec}(A)$ is affine and that $\mathcal{L}$ is trivial. Then $A$ is a Noetherian normal domain and $t$ is an element of its fraction field such that $\mathrm{ord}_{A_\mathfrak{p}}(t) = 0$ for all height 1 primes $\mathfrak{p}$ of $A$. Our goal is to show that $t$ is a unit of $A$. Since $A_\mathfrak{p}$ is a discrete valuation ring for height one primes of $A$ (Algebra, Lemma 031S), the condition signifies that $t \in A_\mathfrak{p}^*$ for all primes $\mathfrak{p}$ of height 1. This implies $t \in A$ and $t^{-1} \in A$ by Algebra, Lemma 031T and the proof is complete. □